

Fairness and AI

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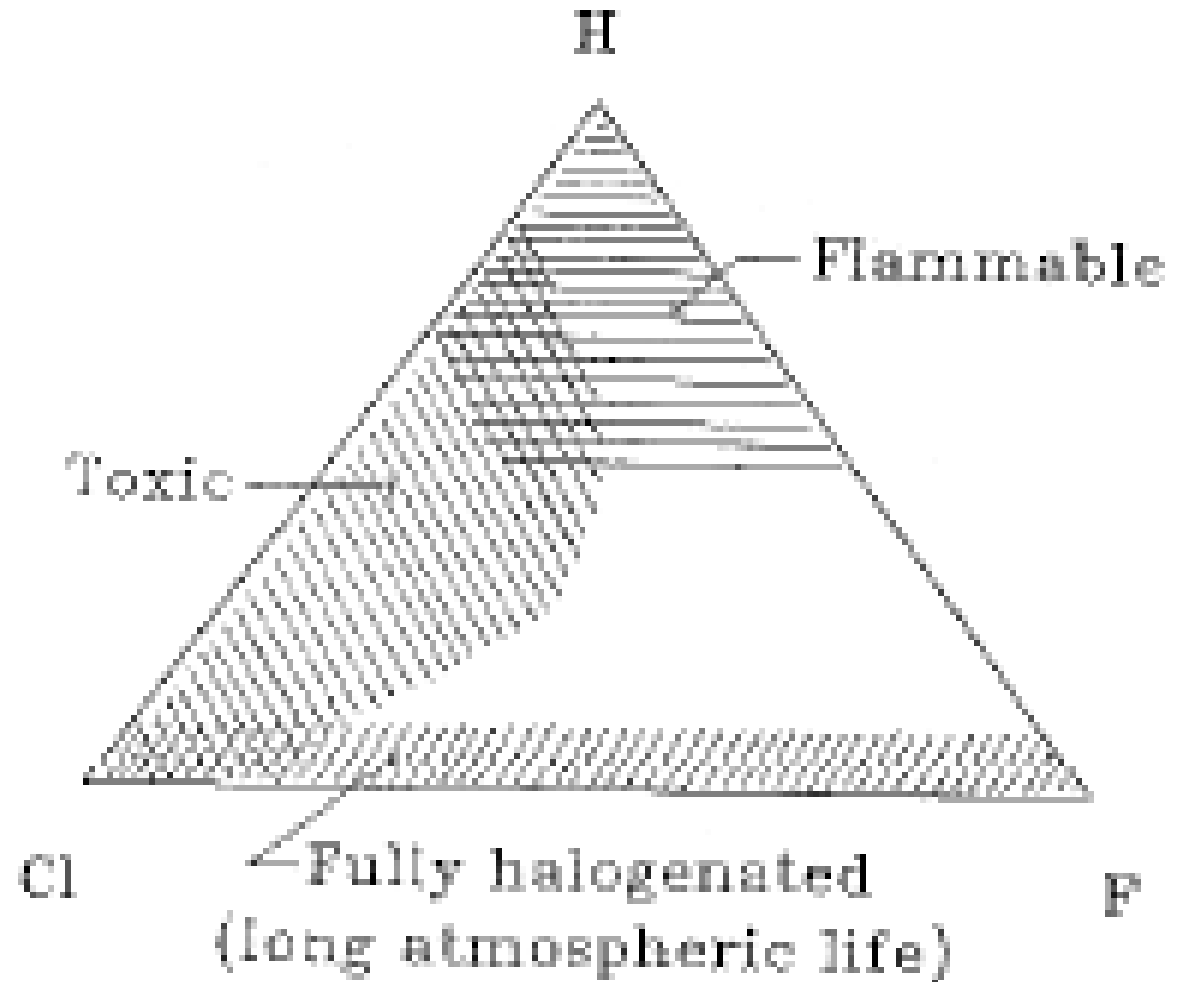
Agenda

- Brief review “Design for Values”
- Algorithmic fairness
- Example of learning analytics
- Relational justice
- Punitive justice

Review “Design for Values”

diagram of ethical tradeoffs in refrigerants

Ibo van de Poel, Investigating Ethical Issues
in Engineering Design, *Science and
Engineering Ethics* 7, 3 (2001): 429-446.



Incorporating ethics into AI applications: some approaches

1. Build the system using ethical rules/ values.
("Whenever you detect a human, stop the robot" [SAFETY]
"Do not make more systematic errors for any sub-group in the dataset compared to the others" [FAIRNESS])
2. Train systems on large data sets in which the rules are followed (e.g., cases where the robot stops for humans)
3. Train systems on large data sets in which the rules might not have been followed, but impose the rules as filters.
4. Rely on the socio-technical system (users) to provide ethical input.

Algorithmic fairness

https://en.wikipedia.org/wiki/Lady_Justice#/media/File:Berner_Iustitia.jpg



Theories of justice

Desert-based (Thrasymachus, Nozick)

Egalitarian (Rawlsian, prioritarian)

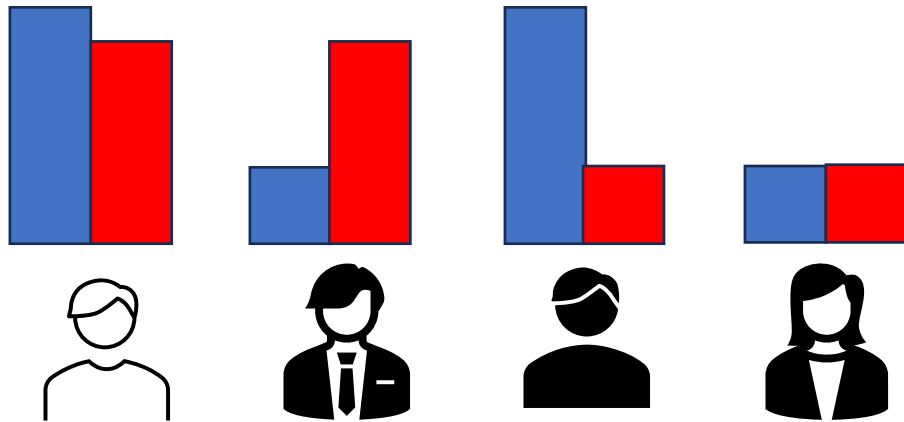
Relational (particularist/ historical)

Procedural/ substantive fairness

Opportunity based/ outcome based

Distributions of benefit

- Current well-being = blue
- Amount of benefit predicted = red



“Each should end up with the same benefit.”

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Desert-based

- Desert = the quality of *being deserved*
- “libertarian” or “meritocratic”
- The assumption is that through talent and effort, people initially acquire basic goods that they deserve. After initial acquisition, fair exchanges cause goods to change hands.
- So long as the initial acquisition and the exchanges are legitimate, the resulting state is regarded as fair.

Justice and mutual agreement

- In the history of Western philosophy, justice has often been identified with the basic rules that people agree to, as the basis for society.
- Contractualism (e.g., Hobbes, Locke): Justice is what people actually agree to in the “state of nature”.
- Contractarianism (e.g., Rawls, Scanlon): Justice is what reasonable people would agree to under ideal circumstances of deliberation.

Rawlsian justice as fairness

- Justice is what reasonable people would agree to as a principle of distributing freedoms/ powers and wealth, if they did not know which position in society they would go on to occupy. (This condition is known as the “Veil of Ignorance”.)
- It results in two main elements:
 - “each person is to have an equal right to the most extensive basic liberty compatible with a similar liberty for others;
 - “social and economic inequalities are to be arranged so that they are both (a) reasonably expected to be to everyone’s advantage, and (b) attached to positions and offices open to all.” (Rawls, *A Theory of Justice*, p. 60)

Competing principles of egalitarianism

- Strict egalitarianism: Intervention (e.g., redistribution) should be used to ensure equal outcomes.
- The Difference Principle: After equal basic liberty and fair institutional equality are established, inequalities are permitted only when they benefit those who are least well off.
- Prioritarianism: Increases in well being for those who are less well-off *count more* than similar increases for those who are better off.

Relational Justice

- Some approaches to justice, instead of affirming a general principle of distribution, focus instead on *power* and *history*.
- In data ethics, biobanking, and research ethics, this has led to many voicing the slogan “Nothing about us, without us” and calls for inclusivity and power-sharing among decision-makers.

HeLa ([/'hi:lɑ:/](#)) is an [immortalized cell line](#) used in scientific research. It is the oldest [human cell line](#) and one of the most commonly used.^{[1][2]} HeLa cells are durable and prolific, allowing for extensive applications in scientific study.^{[3][4]} The line is derived from [cervical cancer](#) cells taken on February 8, 1951,^[5] from [Henrietta Lacks](#), a 31-year-old African American woman, after whom the line is named. Lacks died of cancer on October 4, 1951.^[6]

Relational Justice and Secondary Use

- In the early 2000s, the Havasupai tribe discovered that their DNA samples, which had been collected for studies of diabetes, had been used without consent for other genetic studies. They sued.
- A research study was conducted with IRB chairs and human genetics researchers about the case. “Interviewees perceived no direct impact from the Havasupai case on their work; if they did, it was the perceived need to safeguard themselves by obtaining broad consent or shying away from research with indigenous communities.” (Garrison 2013, 202).

Garrison, Nanibaa'A. "Genomic justice for Native Americans: impact of the Havasupai case on genetic research." *Science, Technology, & Human Values* 38, no. 2 (2013): 201-223.

Procedural fairness vs. substantive fairness (CHLEP p. 80)

“substantive notions and rules directly refer to a particular political or normative goal or outcome a judgment or decision should effectuate”
[where outcomes could also refer to intermediate states such as opportunities or capabilities]

“procedural concepts and rules describe how judgments and decisions and decisions in society should be made” [i.e., what process should be used]

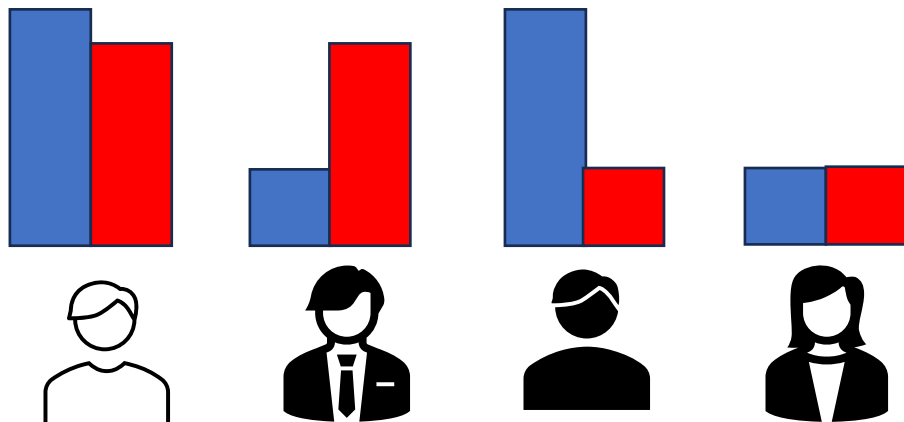
Defining algorithmic fairness

Algorithmic fairness is the property of an algorithm, such that within its social and technical embedding, it embodies an appropriate value of fairness.

See Mitchell et al. 2021 for a discussion of existing approaches and standard examples.

Distributions of benefit

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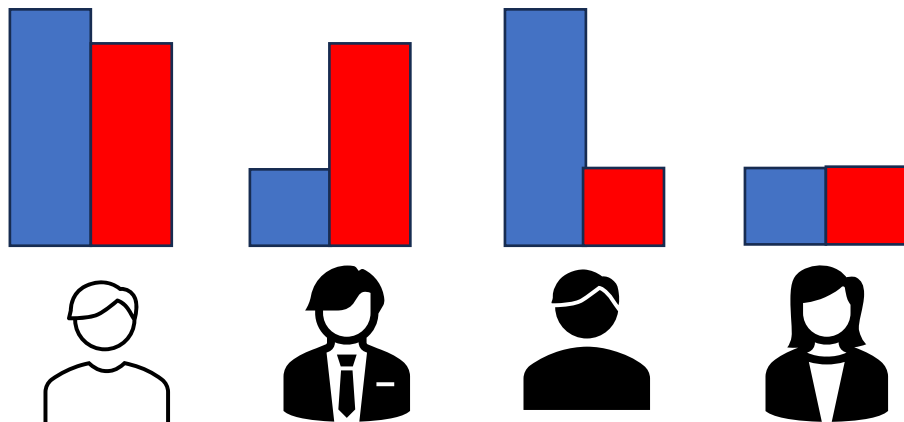


Fairness through intervention. This approach aims to attain algorithmic fairness by directly introducing technical algorithmic interventions either during the model development phase (i.e. in-processing¹) or afterwards as an additional layer (i.e. post-processing²). Within a model development pipeline, in order to detect and mitigate bias, alternative fairness interventions techniques are available and used in practice. Most commonly, to the best of our knowledge, fairness interventions are implemented by means of a *Parity-based approach*, aiming to achieve a form of parity of outcomes between different groups. Below we describe the main idea behind

Barsotti, Flavia, and Rüya Gökhan Koçer.
"MinMax fairness: from Rawlsian Theory of Justice to solution for algorithmic bias." *AI & SOCIETY* 39, no. 3 (2024): 961-974.

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Is algorithmic fairness blind?

1. The elements that make an algorithmic distribution fair must not be accidental. (Follows from robustness.)
2. For the elements to be non-accidental, they must be represented in the algorithm.
3. The elements that make a distribution fair are an individual's prior well-being or need, as well as the change to their condition that is predicted; as well as a representation of parity between groups.
4. Therefore, a fair application must represent all relevant individuals' prior well being or need, the predicted change to their condition, and group membership.

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A worry about 2: What if we conceptualize fairness as *non-bias*? If that is a reasonable conception of fairness, then we don't need to represent the elements that make the distribution fair in the algorithm. Instead, we only need to ensure that there is no bias in the algorithm.

What is (algorithmic) bias?

- The training data does not represent the population to which a model is applied. (Could be due to inclusion, or annotation/ classification.)
“selection bias”
- The model itself introduces incorrect assumptions (e.g., in terms of what input it prioritizes in assigning a risk score). **“model bias”**



Annotation/ classification and bias

task instructions for data collectors and annotators. A typical assignment is illustrated by a data collection project of Active Data: the company received task instructions to collect images of diverse human faces from a Western European company, producing identification and verification systems. Eva, the founder of Active Data, offered more details:

“They were interested in a diversity of five different ethnicities, so Caucasian, African, Middle Eastern, Latin American and Asian. Of course, very debatable whether these can be the five categories that can classify people around the world ”

This type of assignment generally revolves around a client’s envisioned computer vision product and underlying business idea. The technical assumptions of a classification system demand mutually exclusive categories, in this case even for a problematic concept such as race. Whether such categorisation captures the realities of data subjects or coincides with the values and beliefs of data workers is not negotiated. Written instructions formulated by the requester are passed along to project managers who brief workers. Workers then start collecting the images. For outsourcing companies, the rationale behind data-related decisions is “doing what the client ordered” and “offering value to the client.” Conversely,

Milagros Miceli et al., “Documenting computer vision datasets: An invitation to reflexive data practices,” *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency* (ACM, 2021), <https://dl.acm.org/doi/10.1145/3442188.3445880>, p. 165

Problems with fairness as algorithmic non-bias

- Often, when the model achieves non-bias by failing to represent a given characteristic (with the idea that “justice is blind”), background inequalities are left in place.
- Wachter et al. 2021: bias-preserving vs. bias-transforming metrics
- Sometimes, proxies for a characteristic are included in a “blind” model, either accidentally or intentionally. (Barsotti & Koçer 2024, op. cit. conclude that this is a reason to avoid such an approach.)
 - E.g. postal-code and ethnicity, age, or whatever

Example: learning analytics

- The use of data science to support learning
- At TU/e, “data models and algorithms used for learning analytics are explainable, transparent, inclusive, and fair”
- LA may not be used to “[differentiate] between students based on personal background characteristics or any other bias to provide a different quality of education as opposed to other students, discriminate students, or exclude students of educational participation or access to education.”

Learning analytics for TU/e teachers

IDEA

***Think about what you want to achieve,
and how learning analytics can support
your students and/or yourself as a
teacher.***

***Check whether your idea fits with the
TU/e Learning Analytics Code of Practice,
and discuss your ideas with your
departmental teacher supporter.***

Learning analytics at TU/e

PERSONAL DATA USED

-
- Green-listed personal data categories include prior education, background characteristics (age, gender, nationality, preferred language), study and education data, and study behavior.

PERSONAL DATA NOT USED

-
- Special categories of data (e.g., racial or ethnic origin, health, financial data, sexual orientation) can only be processed with explicit consent and approval after review from the ethical committee and privacy team.
-

Punitive Justice

Algorithms are used extensively in some parts of the world to partially automate decisions about whether to keep somebody in jail before they go to trial, or to indicate sentencing guidelines upon conviction for a crime.

Punitive justice follows a somewhat different logic from distributive justice, but it is strongly bound by the principle that “justice is blind” to characteristics of persons such as sex, race, and income. For this reason, fairness in the sense of non-arbitrariness remains a guiding value.

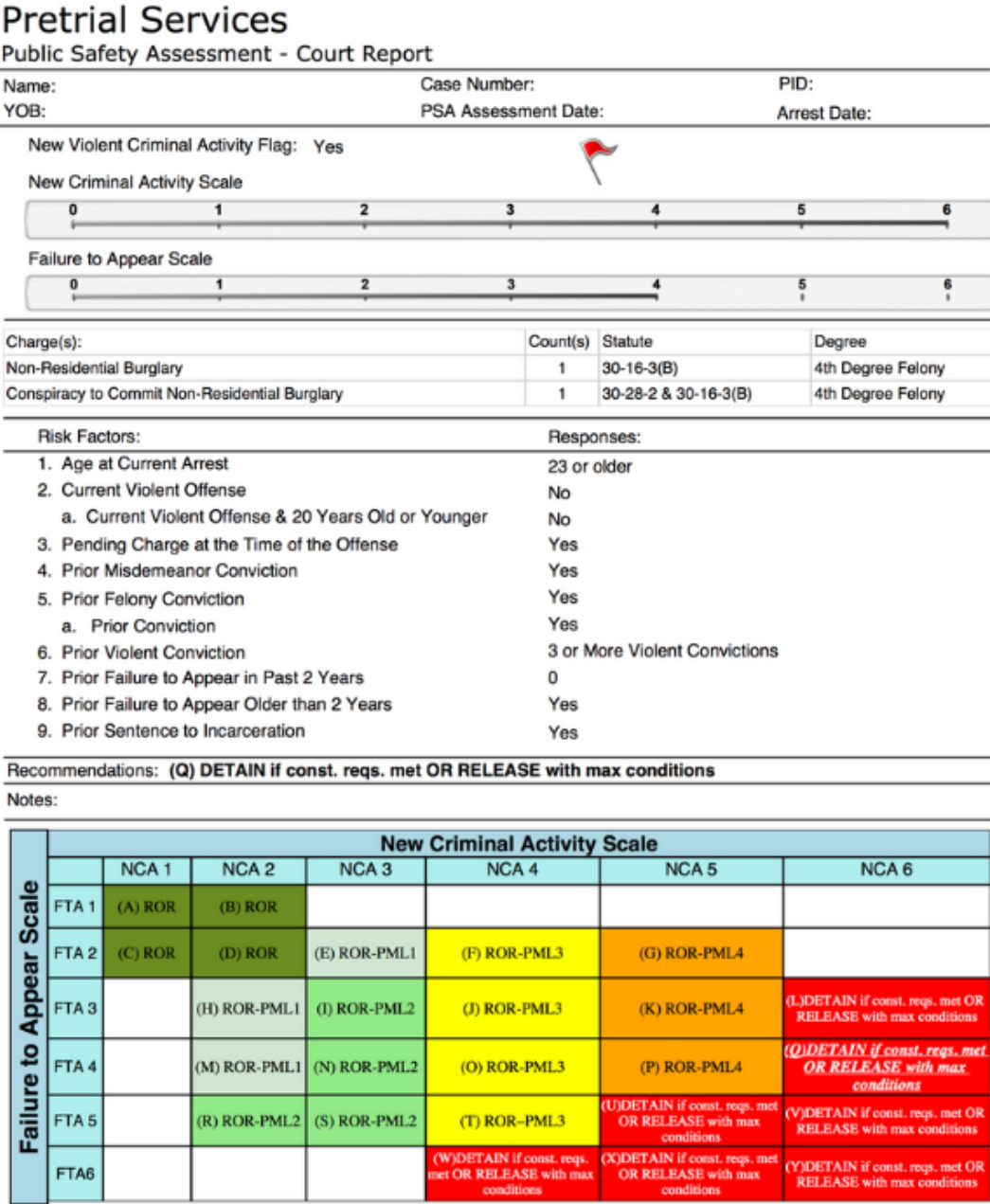


Figure 1

A sample Public Safety Assessment (PSA) report presented to a judge in Bernalillo County (Albuquerque) up through 2022. It shows the defendant’s Failure to Appear (FTA) and

Should we use automated scoring for punitive justice?

“So, should we use the PSA at all? Like Arnold Ventures, we do not support using the PSA to justify detention. We believe that New Mexico’s current system, where the prosecutor must present individualized evidence that a defendant is dangerous, is the right approach from an ethical and constitutional point of view. We should not deny people their liberty on statistical grounds.”

[Pretrial Risk Assessment on the Ground: Algorithms, Judgments, Meaning, and Policy · Summer 2023](#)

Engaging as AI experts

‘If you want to help the real world, dive into it. We often approach interdisciplinary work as a kind of transaction. A collaborator from another field gives us data, and we apply our techniques to it. We each stay in our lane: they are the domain experts, and we are “domain agnostic,” staying aloof from what the data is actually about.

But deep interdisciplinary work means crossing these lanes. If you care about the domain, you should learn about it. If you care about housing and credit, study the history of segregation; if you care about justice, find out how the justice system works. Approach your work with your whole brain and heart. Your life will be more intellectually satisfying, you will grow as a scientist, and your work will be more likely to be useful to the world.’

<https://mit-serc.pubpub.org/pub/czviu6qc/release/2>

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Sandra Wachter, Brent Mittelstadt, and Chris Russell, “Bias preservation in machine learning: The legality of fairness metrics under EU non-discrimination law” (Social Science Research Network, 2021) *SSRN Scholarly Paper* 3792772, <https://papers.ssrn.com/abstract=3792772>, accessed April 28, 2022

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